

Interpretation of KKT Multipliers in Energy System Optimization

Capacity, Dispatch, and Multi-carrier Networks in PyPSA

Alberto Alamia

May 2026

Contents

1	Introduction and Problem Formulation	3
1.1	The Role of Lines and KVL	3
1.2	Generalised Dispatchable Components: Full Problem Formulation	4
2	Lagrangian and KKT Conditions	5
2.1	Lagrangian	5
2.2	Stationarity Conditions	6
2.3	Complementary Slackness	6
2.4	Sign Convention and Non-negativity	7
3	Objective Decomposition: Primal and Dual Perspectives	7
3.1	Primal (Cost) Decomposition	7
3.2	Dual (Revenue) Decomposition	7
3.3	Rents and the General Case	8
4	Role of Binding Constraints	9
4.1	Complementary Slackness and Economic Scarcity	9
4.2	Types of Rent	9
4.3	Zero-Profit Conditions	10
5	Extension: Storage Components	10
5.1	Full Problem Formulation with Storage	10
5.2	KKT Conditions for Storage	11
6	Interpretation of Stationary Conditions	11
6.1	Dispatch: price decomposition	11
6.2	Generator capacity: investment equilibrium	12
6.3	Line capacity: transmission investment equilibrium	12
6.4	Storage: intertemporal arbitrage	12
6.5	Generalised link: full stationarity	12
7	Nodal Marginal Prices	12
8	Examples	13
8.1	Single Node: Baseload and Peaker Competition, LCOP Equivalence	13
8.1.1	Setup	13
8.1.2	KKT Conditions	13
8.1.3	LCOP Equals Energy-Weighted Nodal Price	14
8.2	Three-Node System with Transmission Investment (Triangle)	15
8.2.1	Setup	15
8.2.2	KKT Conditions	15

8.3	Storage Investment with Cycling Constraint	16
8.3.1	Setup	16
8.3.2	KKT Conditions	16
8.4	Multi-carrier Link: One Input, Two Outputs	17
8.4.1	Setup	17
8.4.2	Stationarity Condition	17
8.4.3	Capacity Stationarity	17
8.4.4	Economic Interpretation	18
8.5	Levelized Cost of Production for Multi-carrier Links	18
8.5.1	Motivation	18
8.5.2	LCOP for a Multi-output Link	18
8.5.3	Allocating LCOP to Individual Products	18
8.5.4	LCOP Equals Product-Weighted Nodal Price	19
8.5.5	Competing Links on the Same Bus	19
8.6	CO ₂ Emission Cap via Store	20
8.6.1	Setup	20
8.6.2	Modified Stationarity for Emitting Generators	20
8.6.3	KKT Multipliers of the CO ₂ Store	20
8.6.4	Comparison with a Global Constraint	21
9	Summary: KKT Conditions	21
10	PyPSA Best Practices: Saving KKT Multipliers	21
10.1	Automatically Saved Multipliers	22
10.2	Capacity KKT Multipliers (Extendable Components)	22
10.3	KVL Constraint Duals ($\kappa_{c,t}$)	23
10.4	Cycling Store Constraint Dual	24
10.5	Computing LCOP in PyPSA	24
11	Nomenclature	25

1 Introduction and Problem Formulation

In energy system models we distinguish between two types of decisions:

- **Dispatch decisions:** how much to generate at each time step t
- **Capacity decisions:** how much installed capacity to build (investment)

A **capacity and dispatch** (or *investment and operation*) optimisation jointly determines both [4, 5]. The general linear problem takes the form:

$$\min_{g, G, f, F} \quad \sum_s c_s G_s + \sum_l c_l F_l + \sum_{s,t} w_t o_s g_{s,t} \quad (1)$$

$$\text{s.t.} \quad \sum_{s \in \mathcal{S}_n} g_{s,t} - \sum_{l: n^-(l)=n} f_{l,t} + \sum_{l: n^+(l)=n} f_{l,t} = d_{n,t} \quad \forall n, t \quad (2)$$

$$0 \leq g_{s,t} \leq G_s \quad \forall s, t \quad (3)$$

$$-F_l \leq f_{l,t} \leq F_l \quad \forall l, t \quad (4)$$

$$\sum_{\ell} C_{\ell c} x_{\ell} f_{\ell,t} = 0 \quad \forall c, \forall t \quad (5)$$

$$G_s \geq 0 \quad \forall s, \quad F_l \geq 0 \quad \forall l \quad (6)$$

where:

- $g_{s,t} \geq 0$ is the dispatch of generator s at time t
- $G_s \geq 0$ is the installed capacity of generator s
- $f_{l,t}$ is the power flow on line l at time t
- $F_l \geq 0$ is the capacity of line l
- c_s, c_l are annualised capital costs (€/MW/year)
- o_s is the variable cost (€/MWh)
- w_t is the snapshot weight (hours per snapshot)
- $d_{n,t}$ is the demand at node n and time t
- $C_{\ell c}$ is the cycle incidence matrix, x_{ℓ} is the branch reactance

1.1 The Role of Lines and KVL

The status of line flows $f_{l,t}$ as optimisation variables is a frequent source of confusion and is worth addressing carefully.

Two equivalent perspectives. The LOPF problem admits several equivalent formulations that differ in what variables are used explicitly [1]. The classical *angle formulation* uses the bus voltage angles $\theta_{n,t}$ as the primary variables; the flows are then derived quantities:

$$f_{l,t} = \frac{\theta_{n^-(l),t} - \theta_{n^+(l),t}}{x_l} \quad (7)$$

and KVL is satisfied *automatically* because every cycle is in the kernel of the incidence matrix.

The *Kirchhoff formulation* (used by PyPSA [1]) promotes $f_{l,t}$ to an explicit optimisation variable and enforces KVL as an explicit equality constraint (5):

$$\sum_{\ell \in \text{cycle } c} C_{\ell c} x_{\ell} f_{\ell,t} = 0 \quad \forall c = 1, \dots, L - N + 1, \quad \forall t \quad (8)$$

where $C_{\ell c} \in \{-1, 0, +1\}$ is the cycle incidence matrix and $L - N + 1$ is the number of independent cycles in the network.

Are line flows free optimisation variables? In the Kirchhoff formulation, yes — $f_{l,t}$ are formal optimisation variables, and the KVL constraints (8) are explicit equality constraints. Both perspectives yield *mathematically identical solutions*; the difference is only computational.

The KVL constraints are far from trivial for meshed networks: for a network with N buses and L lines they provide $L - N + 1$ independent equations per snapshot. Together with the KCL nodal balance (2) (which provides $N - 1$ independent equations), they uniquely determine all L flows given the power injections. The “degrees of freedom” for line flows are therefore $L - (N - 1) - (L - N + 1) = 0$: once the injections are set, the physics uniquely fix every flow.

Special case: tree networks. For a network with no cycles ($L = N - 1$, a spanning tree), $L - N + 1 = 0$ and the KVL constraint set is empty. In this case the flows are *uniquely* determined by KCL alone, and the KVL constraint is vacuously satisfied. The triangle example (Section 8.2) illustrates the meshed case.

KVL dual variables. The KVL constraints (5) are equality constraints and therefore have an unrestricted dual variable $\kappa_{c,t}$ (one per independent cycle and snapshot). These appear in the Lagrangian and affect the stationarity condition for the line flows. Their economic interpretation is discussed in Section 2 together with the full set of KKT conditions.

In PyPSA. PyPSA implements the Kirchhoff formulation by default [1], promoting line flows to explicit optimisation variables and adding the KVL equality constraints (8) to the problem. Generators with `p_nom_extendable = True` have G_s as a free variable; lines with `s_nom_extendable = True` have F_l as a free variable.

1.2 Generalised Dispatchable Components: Full Problem Formulation

Many components in multi-energy PyPSA models — electrolyzers, heat pumps, gas turbines, HVDC links, one-way pipelines — are better described as **controllable, dispatchable links with efficiency**: they withdraw a controlled flow from an input bus, convert it, and inject the result into an output bus. This corresponds to the Link component in PyPSA.

A **generalised dispatchable component** s is characterised by:

- An **input bus** $n^-(s)$, from which it withdraws $g_{s,t}$
- An **output bus** $n^+(s)$, into which it injects $\eta_s g_{s,t}$
- An **efficiency** $\eta_s \in (0, 1]$
- A **variable cost** $o_s \geq 0$ per unit of input
- A **capital cost** $c_s \geq 0$ per unit of capacity G_s
- **Monodirectional** dispatch: $g_{s,t} \in [0, G_s]$

The full problem — generators, passive lines, and generalised links — is then:

$$\min_{g, G, f, F} \sum_s c_s G_s + \sum_l c_l F_l + \sum_{s,t} w_t o_s g_{s,t} \quad (\text{P})$$

$$\text{s.t.} \quad \underbrace{\sum_{s: n^+(s)=n} \eta_s g_{s,t} - \sum_{s: n^-(s)=n} g_{s,t} + \sum_l K_{nl} f_{l,t}}_{\text{KCL at node } n} = d_{n,t} \quad \forall n, t \quad (\text{KCL})$$

$$\sum_\ell C_{\ell c} x_\ell f_{\ell,t} = 0 \quad \forall c, \forall t \quad (\text{KVL})$$

$$0 \leq g_{s,t} \leq G_s \quad \forall s, t \quad (\text{D})$$

$$-F_l \leq f_{l,t} \leq F_l \quad \forall l, t \quad (\text{FL})$$

$$G_s \geq 0, \quad F_l \geq 0 \quad (9)$$

This formulation (P)–(FL) is the reference throughout the rest of this document. The extension to storage components is treated separately in Section 5 because storage introduces intertemporal coupling that merits its own analysis.

Table 1 shows how generators and bidirectional links are special cases of the generalised component.

Component	$n^-(s)$	$n^+(s)$	η_s	o_s	Stationarity (interior dispatch)
Generator	(fuel/slack)	electricity bus	≤ 1	> 0	$\lambda_{n,t} = w_t o_s / \eta_s$
Lossless link (one dir.)	bus A	bus B	1	0	$\lambda_{A,t} = \lambda_{B,t}$
Lossy converter	bus A	bus B	< 1	≥ 0	$\lambda_{A,t} = \eta_s \lambda_{B,t} + w_t o_s$
Bidirectional link	two components s^+, s^-		1	0	$\lambda_{A,t} - \lambda_{B,t} = \bar{\mu}_t^+ - \bar{\mu}_t^-$

Table 1: Special cases of the generalised dispatchable component. The last column shows the stationarity condition when the component is freely dispatched ($0 < g_{s,t} < G_s$, so all bound duals vanish). This is the *zero-profit condition*, discussed in Section 4.3.

2 Lagrangian and KKT Conditions

2.1 Lagrangian

We introduce dual variables for each constraint in problem (P)–(FL) following the standard Lagrangian duality framework [2, 3]:

- $\lambda_{n,t}$: dual of the nodal balance (KCL) (equality, unrestricted sign)
- $\kappa_{c,t}$: dual of the KVL constraint (KVL) (equality, unrestricted sign)
- $\underline{\mu}_{s,t} \geq 0$: dual of the lower dispatch bound $g_{s,t} \geq 0$
- $\bar{\mu}_{s,t} \geq 0$: dual of the upper dispatch bound $g_{s,t} \leq G_s$
- $\bar{\mu}_{l,t}^+ \geq 0$: dual of the upper flow bound $f_{l,t} \leq F_l$
- $\bar{\mu}_{l,t}^- \geq 0$: dual of the lower flow bound $f_{l,t} \geq -F_l$

We assume $G_s > 0$ and $F_l > 0$ at optimality, so the non-negativity duals for capacities are zero by complementary slackness and are omitted.

The Lagrangian of problem (P)–(FL) is:

$$\begin{aligned}
\mathcal{L} = & \sum_s c_s G_s + \sum_l c_l F_l + \sum_{s,t} w_t o_s g_{s,t} \\
& - \sum_{n,t} \lambda_{n,t} \left(\sum_{s: n^+(s)=n} \eta_s g_{s,t} - \sum_{s: n^-(s)=n} g_{s,t} + \sum_l K_{nl} f_{l,t} - d_{n,t} \right) \\
& - \sum_{c,t} \kappa_{c,t} \left(\sum_\ell C_{\ell c} x_\ell f_{\ell,t} \right) \\
& - \sum_{s,t} \underline{\mu}_{s,t} g_{s,t} + \sum_{s,t} \bar{\mu}_{s,t} (g_{s,t} - G_s) \\
& + \sum_{l,t} \bar{\mu}_{l,t}^+ (f_{l,t} - F_l) - \sum_{l,t} \bar{\mu}_{l,t}^- (f_{l,t} + F_l)
\end{aligned} \tag{10}$$

The stationarity conditions below follow directly from setting $\partial \mathcal{L} / \partial x = 0$ for each decision variable x .

2.2 Stationarity Conditions

With respect to dispatch $g_{s,t}$:

$$\boxed{w_t o_s - \lambda_{n^+(s),t} \eta_s + \lambda_{n^-(s),t} - \underline{\mu}_{s,t} + \bar{\mu}_{s,t} = 0} \quad (11)$$

For a generator injecting into bus n (with $n^-(s)$ a virtual fuel bus at zero price):

$$\lambda_{n,t} = w_t o_s + \bar{\mu}_{s,t} - \underline{\mu}_{s,t} \quad (12)$$

With respect to generator capacity G_s :

$$\boxed{c_s = \sum_t \bar{\mu}_{s,t}} \quad (13)$$

With respect to line flow $f_{l,t}$: In the Kirchhoff formulation $f_{l,t}$ is a decision variable, so a stationarity condition exists:

$$\frac{\partial \mathcal{L}}{\partial f_{l,t}} = 0 \implies \boxed{-\lambda_{n^+(l),t} + \lambda_{n^-(l),t} - \sum_c \kappa_{c,t} C_{lc} x_l + \bar{\mu}_{l,t}^+ - \bar{\mu}_{l,t}^- = 0} \quad (14)$$

Interpretation. For an uncongested line ($\bar{\mu}_{l,t}^+ = \bar{\mu}_{l,t}^- = 0$):

$$\lambda_{n^-(l),t} - \lambda_{n^+(l),t} = \sum_c \kappa_{c,t} C_{lc} x_l \quad (15)$$

The nodal price difference across an uncongested line is *not* zero in a meshed network: it equals the reactance-weighted contribution of the KVL duals $\kappa_{c,t}$. This is the mathematical expression of Kirchhoff's physics — the price signal propagates through the network topology. Only in a radial (tree) network, where there are no cycles and $\kappa_{c,t}$ vanishes identically, does an uncongested line guarantee equal prices at both ends.

With respect to line capacity F_l :

$$\boxed{c_l = \sum_t (\bar{\mu}_{l,t}^+ + \bar{\mu}_{l,t}^-)} \quad (16)$$

2.3 Complementary Slackness

For each inequality constraint, complementary slackness states that the dual variable is non-zero *only* when the constraint is active (binding):

$$\underline{\mu}_{s,t} \geq 0, \quad \underline{\mu}_{s,t} \cdot g_{s,t} = 0 \implies \underline{\mu}_{s,t} > 0 \iff g_{s,t} = 0 \quad (17)$$

$$\bar{\mu}_{s,t} \geq 0, \quad \bar{\mu}_{s,t} \cdot (g_{s,t} - G_s) = 0 \implies \bar{\mu}_{s,t} > 0 \iff g_{s,t} = G_s \quad (18)$$

$$\bar{\mu}_{l,t}^+ \geq 0, \quad \bar{\mu}_{l,t}^+ \cdot (f_{l,t} - F_l) = 0 \implies \bar{\mu}_{l,t}^+ > 0 \iff f_{l,t} = F_l \quad (19)$$

$$\bar{\mu}_{l,t}^- \geq 0, \quad \bar{\mu}_{l,t}^- \cdot (f_{l,t} + F_l) = 0 \implies \bar{\mu}_{l,t}^- > 0 \iff f_{l,t} = -F_l \quad (20)$$

The key consequence: a dual variable can only be non-zero when its constraint is *tight*. Slack constraints contribute nothing to the stationarity conditions and carry zero economic value.

2.4 Sign Convention and Non-negativity

For the KKT conditions to be necessary and sufficient (strong duality), the problem must be convex, which is guaranteed here (linear objective and linear constraints). The sign conventions are:

KKT Sign Summary

- $\lambda_{n,t}$: unrestricted (equality constraint dual); typically positive as it represents the value of one additional MW of supply at node n
- $\kappa_{c,t}$: unrestricted (equality constraint dual)
- $\underline{\mu}_{s,t} \geq 0$: non-negative (lower bound dual); positive when generator is idle
- $\bar{\mu}_{s,t} \geq 0$: non-negative (upper bound dual); positive when generator is at full capacity
- $\bar{\mu}_{l,t}^{\pm} \geq 0$: non-negative (flow bound duals); positive when line is congested

3 Objective Decomposition: Primal and Dual Perspectives

The objective function of a linear programme is a sum over all decision variables weighted by their cost coefficients. This has an immediate consequence: the optimal total cost can be partitioned *exactly* into contributions from any non-overlapping grouping of the variables. Applied to an energy system, this allows the total system cost to be attributed to individual plants, technology classes, carriers, or geographic zones — with no approximation. Two complementary decompositions are possible: one in terms of primal variables (direct costs), one in terms of dual variables (nodal prices and rents). They give the same number per group, but carry different economic interpretations.

3.1 Primal (Cost) Decomposition

Partitioning all components into non-overlapping subsets $\mathcal{P}_1, \mathcal{P}_2, \dots$ (each component belongs to exactly one subset; together they cover all components), the objective (P) decomposes by linearity of the sum:

$$\text{obj}^* = \sum_k \underbrace{\sum_{s \in \mathcal{P}_k} \left(c_s G_s^* + \sum_t w_t o_s g_{s,t}^* \right)}_{\text{cost}(\mathcal{P}_k)} \quad (21)$$

This is the **engineering (accounting) decomposition**: each subset's contribution is the sum of its capital expenditure (annualised investment cost $c_s G_s^*$) and operational expenditure ($w_t o_s g_{s,t}^*$, fuel and variable costs). The partition can be drawn at any level of granularity, and the subset totals always sum exactly to the system cost. No knowledge of dual variables is required.

3.2 Dual (Revenue) Decomposition

At LP optimality the KKT conditions derived in Section 2 reveal an equivalent representation of every component's total cost in terms of the nodal prices $\lambda_{n,t}$:

Theorem: Dual Objective Decomposition

For any **extendable** component s with $G_s^* > 0$ at LP optimality, the KKT conditions imply:

$$\underbrace{c_s G_s^* + \sum_t w_t o_s g_{s,t}^*}_{\text{total cost of } s} = \underbrace{\sum_t w_t \lambda_{n(s),t} g_{s,t}^*}_{\text{revenue of } s \text{ at nodal prices}} \quad (22)$$

Consequently, for a system where all capacity is extendable:

$$\text{obj}^* = \sum_s \sum_t w_t \lambda_{n(s),t} g_{s,t}^* = \sum_k \underbrace{\sum_{s \in \mathcal{P}_k} \sum_t w_t \lambda_{n(s),t} g_{s,t}^*}_{\text{revenue}(\mathcal{P}_k)} \quad (23)$$

The same partition $\mathcal{P}_k$ used in the primal decomposition (21) gives the same group totals via the dual formula.

Proof sketch. Three KKT conditions combine to yield (22):

- **Investment equilibrium** (13): $c_s = \sum_t \bar{\mu}_{s,t}$, so $c_s G_s^* = \sum_t \bar{\mu}_{s,t} G_s^*$.
- **Complementary slackness** (18): $\bar{\mu}_{s,t} > 0$ forces $g_{s,t}^* = G_s^*$, hence $\bar{\mu}_{s,t} G_s^* = \bar{\mu}_{s,t} g_{s,t}^*$ always holds.
- **Dispatch stationarity** (12): $\lambda_{n,t} = w_t o_s + \bar{\mu}_{s,t}$ whenever $g_{s,t}^* > 0$ (complementary slackness (17) sets $\underline{\mu}_{s,t} = 0$ there).

Combining:

$$c_s G_s^* + \sum_t w_t o_s g_{s,t}^* = \sum_t \bar{\mu}_{s,t} g_{s,t}^* + \sum_t w_t o_s g_{s,t}^* = \sum_t (\bar{\mu}_{s,t} + w_t o_s) g_{s,t}^* = \sum_t \lambda_{n,t} g_{s,t}^* \quad \square$$

The result extends to generalised links, multi-carrier links, and lines via analogous stationarity conditions (Sections 8.4–8.5). A complete proof with numerical illustration is in Section 8.1.3.

3.3 Rents and the General Case

For a **fixed-capacity** component (G_s is a given parameter, not an optimisation variable), the investment equilibrium does not exist: no stationarity condition with respect to G_s is imposed. Dispatch stationarity and complementary slackness still hold, giving:

$$\sum_t w_t o_s g_{s,t}^* = \sum_t w_t \lambda_{n(s),t} g_{s,t}^* - \underbrace{\sum_t \bar{\mu}_{s,t} G_s}_{r_s \geq 0} \quad (24)$$

The term $r_s = \sum_t \bar{\mu}_{s,t} G_s \geq 0$ is the **infra-marginal rent**: the excess of market revenue over operating cost for a fixed-capacity unit that runs below the price-setting marginal cost. For the general mixed system (some extendable, some fixed):

$$\text{obj}^* = \sum_s \sum_t w_t \lambda_{n(s),t} g_{s,t}^* - \sum_{s \text{ fixed}} r_s \quad (25)$$

Two views of the same total cost

- **Primal view** — “**what each plant spends**”: operational cost plus capital repayment. Always exact; requires only primal variables.
- **Dual view** — “**what the market pays each plant**”: revenue at nodal prices $\lambda_{n,t}$, minus the infra-marginal rent retained by fixed-capacity owners. Requires LP optimality.
- **Extendable components** (long-run equilibrium): the two views coincide — zero-profit. The competitive market price signal $\lambda_{n,t}$ exactly recovers each component’s total cost.
- **Fixed-capacity components**: market revenue exceeds operating cost by the infra-marginal rent r_s . This rent is invisible in the primal decomposition but is the primary signal driving new investment decisions.

Connection to the KKT conditions. The dual decomposition (23) holds *because* the KKT conditions are satisfied at optimality. The nodal prices $\lambda_{n,t}$, dispatch rents $\bar{\mu}_{s,t}$, congestion rents $\bar{\mu}_{l,t}^\pm$, and intertemporal storage values $\xi_{k,t}$ are the mechanism that makes the dual revenue equal the primal cost, exactly, for every optimally-invested component. Understanding *which constraints bind* — and therefore which multipliers are non-zero — is the key to understanding the cost structure, investment incentives, and price formation in the optimised system. The rest of this document interprets the KKT multipliers for each component type and shows what each one signals economically.

4 Role of Binding Constraints

4.1 Complementary Slackness and Economic Scarcity

Complementary slackness (17)–(20) gives a precise condition: a dual variable can only be non-zero when its associated constraint holds with equality (is **binding**).

- **Not binding** (slack): dual is zero, no contribution to price formation.
- **Binding**: dual ≥ 0 , actively shapes prices and investment signals.

A binding constraint signals that the system is *scarce* in the resource governed by that constraint. The dual variable measures exactly how much the total system cost would decrease if that constraint were relaxed by one unit — the **shadow price** or **rent** of that constraint.

4.2 Types of Rent

Infra-marginal rent ($\bar{\mu}_{s,t}$). A generator running at full capacity when a more expensive unit sets the price earns a rent $\bar{\mu}_{s,t} = \lambda_{n,t} - w_t o_s > 0$. The system would like more output from this cheap unit but cannot get it (capacity binding).

Congestion rent ($\bar{\mu}_{l,t}^\pm$). A congested line earns a rent equal to the nodal price difference between its endpoints. This rent measures the value of moving one additional MW across the congested path.

Intertemporal rent (ξ_t). Storage units earn a rent by shifting cheap energy to expensive hours. This intertemporal rent propagates across snapshots through the state-of-energy evolution constraint (see Section 5.2).

4.3 Zero-Profit Conditions

The zero-profit condition is the stationarity equation (11) evaluated when neither bound is binding ($\bar{\mu}_{s,t} = \underline{\mu}_{s,t} = 0$):

$$\lambda_{n^-(s),t} = \eta_s \lambda_{n^+(s),t} + w_t o_s \quad (26)$$

Interpretation: the value of the input equals the efficiency-weighted value of the output plus variable cost. A freely dispatched converter earns exactly zero profit: revenue equals production cost.

Any deviation from (26) is evidence of a binding constraint:

- $\lambda_{n^+(s),t} > \eta_s^{-1}(\lambda_{n^-(s),t} - w_t o_s)$: output price exceeds cost — capacity binding ($\bar{\mu}_{s,t} > 0$, earning infra-marginal rent)
- $\lambda_{n^+(s),t} < \eta_s^{-1}(\lambda_{n^-(s),t} - w_t o_s)$: output price below cost — unit idle ($\underline{\mu}_{s,t} > 0$)

Investment equilibrium. Combining (13) with complementary slackness and (12):

$$c_s = \sum_t \bar{\mu}_{s,t} = \sum_{t: g_{s,t}=G_s} (\lambda_{n,t} - w_t o_s) \quad (27)$$

Capital costs are recovered through the *accumulation of infra-marginal rents* at binding hours. No single snapshot need recover the full capital cost.

5 Extension: Storage Components

5.1 Full Problem Formulation with Storage

A **store** k is characterised by a state of energy $e_{k,t}$, charge/discharge flows $g_{k,t}^+$ and $g_{k,t}^-$, efficiencies η_k^+ , η_k^- , power capacity G_k^s , and energy capacity $E_{k,\max}^s$.

The full problem extending (P) with stores is:

$$\begin{aligned} \min \quad & \sum_s c_s G_s + \sum_l c_l F_l + \sum_k (c_k^P G_k^s + c_k^E E_{k,\max}^s) + \sum_{s,t} w_t o_s g_{s,t} \quad (\text{P}') \\ \text{s.t.} \quad & \text{(KCL)} - \text{(FL)} \quad (\text{generators, links, lines as before, with storage contributing to KCL}) \\ & e_{k,t} = e_{k,t-1} + w_t (\eta_k^+ g_{k,t}^+ - \frac{1}{\eta_k^-} g_{k,t}^-) \quad \forall k, t \quad (\text{SOE}) \\ & 0 \leq g_{k,t}^\pm \leq G_k^s \quad \forall k, t \quad (\text{PL}) \\ & 0 \leq e_{k,t} \leq E_{k,\max}^s \quad \forall k, t \quad (\text{EL}) \\ & e_{k,0} = e_{k,T} \quad \forall k \quad (\text{CYC}) \end{aligned}$$

The **cycling constraint (CYC)** requires the state of energy to be the same at the start and end of the optimisation horizon. This is essential for long-run planning models where the store must close the energy balance over the full year: without it, the optimiser would trivially fill the store at the start and drain it at the end, artificially reducing costs.

Dual variables for storage:

- $\xi_{k,t}$: dual of the SOE evolution (SOE) — shadow price of stored energy
- $\underline{\mu}_{k,t}^+, \bar{\mu}_{k,t}^+ \geq 0$: duals of charge power bounds
- $\underline{\mu}_{k,t}^-, \bar{\mu}_{k,t}^- \geq 0$: duals of discharge power bounds
- $\bar{\gamma}_{k,t} \geq 0$: dual of the energy upper bound $e_{k,t} \leq E_{k,\max}^s$
- $\underline{\gamma}_{k,t} \geq 0$: dual of $e_{k,t} \geq 0$

5.2 KKT Conditions for Storage

Stationarity with respect to charging $g_{k,t}^+$:

$$-w_t \eta_k^+ \lambda_{n_k,t} + \bar{\mu}_{k,t}^+ - \underline{\mu}_{k,t}^+ + \xi_{k,t} = 0 \quad (28)$$

Stationarity with respect to discharging $g_{k,t}^-$:

$$\frac{w_t \lambda_{n_k,t}}{\eta_k^-} + \bar{\mu}_{k,t}^- - \underline{\mu}_{k,t}^- - \xi_{k,t} = 0 \quad (29)$$

Here $\xi_{k,t}$ is the shadow price of stored energy at time t .

Intertemporal propagation of $\xi_{k,t}$: The SOE constraint (SOE) links consecutive time steps, giving:

$$\xi_{k,t} = \xi_{k,t+1} + \bar{\gamma}_{k,t} - \underline{\gamma}_{k,t} \quad (30)$$

The shadow price of stored energy at t equals the shadow price one period ahead, adjusted by the energy capacity dual. When the reservoir is neither full nor empty ($\bar{\gamma}_{k,t} = \underline{\gamma}_{k,t} = 0$), $\xi_{k,t} = \xi_{k,t+1}$: the value of an extra unit of stored energy is constant over that period.

Combining charge and discharge stationarity. When storage is actively cycling (neither power constraint binding):

$$w_{t^-} \lambda_{n_k,t^-} = \frac{\xi_{k,t^-}}{\eta_k^-} \quad \text{and} \quad w_{t^+} \lambda_{n_k,t^+} = \frac{\xi_{k,t^+}}{\eta_k^+} \quad (31)$$

Combining, an ideal round trip from charging hour t^+ to discharging hour t^- :

$$\lambda_{n_k,t^-} \geq \frac{\lambda_{n_k,t^+}}{\eta_k^+ \eta_k^-} \quad (32)$$

Storage earns an intertemporal rent by arbitraging between cheap and expensive hours. The round-trip efficiency $\eta_k^+ \eta_k^- < 1$ sets the minimum price ratio required for profitable cycling.

Capacity stationarity for power G_k^s :

$$c_k^P = \sum_t (\bar{\mu}_{k,t}^+ + \bar{\mu}_{k,t}^-) \quad (33)$$

Capacity stationarity for energy $E_{k,\max}^s$:

$$c_k^E = \sum_t \bar{\gamma}_{k,t} \quad (34)$$

6 Interpretation of Stationary Conditions

6.1 Dispatch: price decomposition

From the stationarity condition (12), the nodal price decomposes into variable cost, infra-marginal rent, and below-cost shadow:

- $0 < g_{s,t} < G_s$: $\bar{\mu}_{s,t} = \underline{\mu}_{s,t} = 0 \Rightarrow \lambda_{n,t} = w_t o_s$ (marginal unit; zero rent)
- $g_{s,t} = G_s$: $\bar{\mu}_{s,t} > 0 \Rightarrow \lambda_{n,t} > w_t o_s$ (infra-marginal rent)
- $g_{s,t} = 0$: $\underline{\mu}_{s,t} > 0 \Rightarrow \lambda_{n,t} < w_t o_s$ (too expensive)

6.2 Generator capacity: investment equilibrium

From the investment equilibrium (27), capital costs are recovered through infra-marginal rents at binding hours: $c_s = \sum_{t: g_{s,t}=G_s} (\lambda_{n,t} - w_t o_s)$.

6.3 Line capacity: transmission investment equilibrium

From (16), the line capital cost equals the sum of congestion rents over all snapshots. The congestion rent $\bar{\mu}_{l,t}^+$ measures the nodal price difference across the congested line. In a two-node setting (or radial network): $\bar{\mu}_{l,t}^+ = \lambda_{n^-(l),t} - \lambda_{n^+(l),t}$ when $f_{l,t} = F_l$.

6.4 Storage: intertemporal arbitrage

From (33) and (34):

- Power capacity is sized so that total binding-hour power rents equal c_k^P
- Energy capacity is sized so that total full-reservoir rents equal c_k^E
- The value of stored energy $\xi_{k,t}$ propagates across time via (30)

6.5 Generalised link: full stationarity

From (11) (and the zero-profit form (26)), the input price equals the efficiency-weighted output price plus variable cost plus rent. When freely dispatched ($\bar{\mu}_{s,t} = \underline{\mu}_{s,t} = 0$) this reduces to the zero-profit condition (26).

7 Nodal Marginal Prices

The dual $\lambda_{n,t}$ of the nodal balance constraint is the **Locational Marginal Price (LMP)** at node n and snapshot t [4, 6]:

$$\lambda_{n,t} = \frac{\partial \text{Total Cost}^*}{\partial d_{n,t}} \quad (35)$$

It is the marginal cost of supplying one additional MW of demand at node n at time t .

Generic bus. Consider a bus n with:

- Generators $s \in \mathcal{S}_n$ (injecting)
- Demand $d_{n,t}$ (withdrawing)
- Outgoing links $s \in \mathcal{L}_n^{\text{out}}$ (withdrawing $g_{s,t}$ from bus n)
- Incoming links $s \in \mathcal{L}_n^{\text{in}}$ (injecting $\eta_s g_{s,t}$ into bus n)
- Passive line connections (net flow $\sum_l K_{nl} f_{l,t}$)
- Storage units $k \in \mathcal{K}_n$ (charging $-g_{k,t}^+$, discharging $+g_{k,t}^-$)

The nodal balance at n is:

$$\sum_{s \in \mathcal{S}_n} g_{s,t} + \sum_{k \in \mathcal{K}_n} (g_{k,t}^- - g_{k,t}^+) + \sum_{s \in \mathcal{L}_n^{\text{in}}} \eta_s g_{s,t} - \sum_{s \in \mathcal{L}_n^{\text{out}}} g_{s,t} + \sum_l K_{nl} f_{l,t} = d_{n,t} \quad (36)$$

The LMP $\lambda_{n,t}$ is the shadow price of this constraint and aggregates information from *all* components connected to the bus.

What $\lambda_{n,t}$ reflects.

- **Short-run marginal cost:** variable cost of the marginal generator
- **Infra-marginal rents:** upward pressure from units at full capacity
- **Congestion rents:** price wedge from binding line flow constraints
- **Intertemporal rents:** value of stored energy from storage
- **KVL coupling:** in meshed networks, even uncongested lines contribute to price differences via the KVL duals $\kappa_{c,t}$

Long-run equilibrium. At long-run equilibrium, each invested technology satisfies its investment equilibrium condition. The system-wide price signal $\lambda_{n,t}$ integrates to recover all capital costs — no investment earns more or less than its annualised cost in expectation over the optimisation horizon.

In PyPSA. Nodal marginal prices are accessible as `n.buses_t.marginal_price` after solving. Each row is a snapshot, each column is a bus.

8 Examples

8.1 Single Node: Baseload and Peaker Competition, LCOP Equivalence

8.1.1 Setup

A single bus with two extendable generators:

- Generator 1 (baseload): high capital cost c_1 , low variable cost o_1
- Generator 2 (peaker): low capital cost c_2 , high variable cost o_2

with $o_1 < o_2$ and $c_1 > c_2$.

$$\min \quad c_1 G_1 + c_2 G_2 + \sum_t w_t (o_1 g_{1,t} + o_2 g_{2,t}) \quad (37)$$

$$\text{s.t.} \quad g_{1,t} + g_{2,t} = d_t \quad \forall t \quad (38)$$

$$0 \leq g_{s,t} \leq G_s \quad \forall s, t \quad (39)$$

8.1.2 KKT Conditions

Dispatch stationarity.

- Off-peak: generator 1 at capacity ($g_{1,t} = G_1$), generator 2 partially loaded: $\lambda_t = w_t o_2$ (peaker sets the price). Generator 1 earns rent $\bar{\mu}_{1,t} = \lambda_t - w_t o_1 = w_t(o_2 - o_1) > 0$.
- Peak: generator 2 also at capacity, price may be pushed higher by scarcity.
- When generator 2 is the marginal unit ($0 < g_{2,t} < G_2$, $\bar{\mu}_{2,t} = 0$): $\lambda_t = w_t o_2$.

Capacity stationarity for generator 1:

$$c_1 = \sum_t \bar{\mu}_{1,t} = \sum_{t: g_{1,t}=G_1} (\lambda_t - w_t o_1) = \sum_{t: g_{1,t}=G_1} w_t (o_2 - o_1) \quad (40)$$

Generator 1 recovers capital cost through the price spread $o_2 - o_1$ earned whenever it runs at full capacity while the peaker sets the price.

Capacity stationarity for generator 2:

$$c_2 = \sum_{t: g_{2,t} = G_2} (\lambda_t - w_t o_2) \quad (41)$$

The peaker is at full capacity only during the highest-demand hours, when both generators are fully loaded and no unit has slack. In those **scarcity hours** the price rises above $w_t o_2$ — there is no freely dispatched unit to pin the price at its variable cost — and the peaker earns the difference $\lambda_t - w_t o_2 > 0$ as an infra-marginal rent. Summed over the (typically few) scarcity snapshots, these rents must exactly cover c_2 . A peaker with low capital cost therefore does not need to run continuously; it recovers its investment through infrequent but high-priced events.

8.1.3 LCOP Equals Energy-Weighted Nodal Price

Result: LCOP = energy-weighted LMP

For any optimally invested generator s (with $G_s > 0$):

$$\text{LCOP}_s = \frac{c_s G_s + \sum_t w_t o_s g_{s,t}}{\sum_t w_t g_{s,t}} = \frac{\sum_t w_t g_{s,t} \pi_{n,t}}{\sum_t w_t g_{s,t}} \equiv \bar{\pi}_{n,s}^{\text{energy}} \quad (42)$$

where $\pi_{n,t} = \lambda_{n,t}/w_t$ is the nodal price in €/MWh.

Proof. Step 1: expand the numerator of LCOP using the investment equilibrium condition (13):

$$c_s G_s = \left(\sum_t \bar{\mu}_{s,t} \right) G_s \quad (43)$$

By complementary slackness (18), $\bar{\mu}_{s,t} > 0$ implies $g_{s,t} = G_s$, so:

$$c_s G_s = \sum_t \bar{\mu}_{s,t} G_s = \sum_t \bar{\mu}_{s,t} g_{s,t} \quad (44)$$

Step 2: substitute into the LCOP numerator:

$$c_s G_s + \sum_t w_t o_s g_{s,t} = \sum_t (\bar{\mu}_{s,t} + w_t o_s) g_{s,t} \quad (45)$$

Step 3: apply the dispatch stationarity (12). When $g_{s,t} > 0$, complementary slackness gives $\bar{\mu}_{s,t} = 0$, so:

$$\bar{\mu}_{s,t} + w_t o_s = \lambda_{n,t} \quad \text{whenever } g_{s,t} > 0 \quad (46)$$

Step 4: since only snapshots with $g_{s,t} > 0$ contribute to the sum:

$$c_s G_s + \sum_t w_t o_s g_{s,t} = \sum_t \lambda_{n,t} g_{s,t} = \sum_t w_t g_{s,t} \pi_{n,t} \quad (47)$$

Dividing by $\sum_t w_t g_{s,t}$ gives (42). $\square$

Scope of the result

The result holds for any extendable component ($G_s > 0$, $c_s > 0$) — generator, link, multi-carrier link — as long as the LP is at optimality. It does *not* hold for fixed-capacity components (no investment equilibrium condition exists) or components not invested ($G_s = 0$).

Individual zero-profit condition for each generator. Result (42) holds for each optimally invested generator *independently*: the formula is an individual break-even condition, not a statement that competing generators share the same LCOP.

In the baseload/peaker example, the two LCOPs are generally *different*: the baseload generator produces in cheap off-peak hours (facing lower average prices) while the peaker produces only at high-price peak hours (facing higher average prices). Each breaks even in its own niche — neither needs to match the other’s LCOP. This coexistence is possible precisely because they serve different time periods: the baseload would not cover the peak, and the peaker would be too expensive for continuous operation.

LCOP vs. production-weighted average of $\pi_{n,t}$. The result (42) already answers this: the LCOP and the production-weighted average nodal price are **the same quantity** at optimum — that equality is exactly what was proved. In other words, if you compute $\sum_t w_t g_{s,t} \pi_{n,t} / \sum_t w_t g_{s,t}$ directly from PyPSA output, you obtain the LCOP.

The reason individual snapshots can still have $\pi_{n,t}$ above or below the LCOP is that the weighting by $g_{s,t}$ does the work: snapshots where s is at full capacity ($g_{s,t} = G_s$, $\pi_{n,t}$ above marginal cost) and snapshots where s is the marginal unit ($\pi_{n,t} = w_t o_s$) all enter the average, and the investment equilibrium (13) guarantees they balance out to the LCOP exactly.

8.2 Three-Node System with Transmission Investment (Triangle)

8.2.1 Setup

Three buses A, B, C connected by three extendable lines forming a triangle (the simplest meshed network). $N = 3$, $L = 3$, so there is $L - N + 1 = 1$ independent cycle.

Label the lines: l_1 (A–B), l_2 (B–C), l_3 (A–C). One generator per bus with variable costs $o_A < o_B < o_C$.

KVL constraint (one per snapshot). Choosing the triangle as the independent cycle, with consistent orientation:

$$x_1 f_{1,t} + x_2 f_{2,t} - x_3 f_{3,t} = 0 \quad \forall t \quad (48)$$

where x_i is the reactance of line l_i . The sign of $C_{3c} = -1$ reflects that l_3 is traversed in reverse by the chosen cycle orientation.

Nodal balances.

$$\text{Node A: } g_{A,t} - f_{1,t} - f_{3,t} = d_{A,t} \quad (49)$$

$$\text{Node B: } g_{B,t} + f_{1,t} - f_{2,t} = d_{B,t} \quad (50)$$

$$\text{Node C: } g_{C,t} + f_{2,t} + f_{3,t} = d_{C,t} \quad (51)$$

(Using incidence matrix sign convention: +1 if line ends at bus, −1 if line starts at bus.)

8.2.2 KKT Conditions

Stationarity for line flows. From (14) applied to each line, with the single cycle dual κ_t :

$$l_1 (A-B): \quad \lambda_{A,t} - \lambda_{B,t} = \kappa_t x_1 + \bar{\mu}_{1,t}^+ - \bar{\mu}_{1,t}^- \quad (52)$$

$$l_2 (B-C): \quad \lambda_{B,t} - \lambda_{C,t} = \kappa_t x_2 + \bar{\mu}_{2,t}^+ - \bar{\mu}_{2,t}^- \quad (53)$$

$$l_3 (A-C): \quad \lambda_{A,t} - \lambda_{C,t} = -\kappa_t x_3 + \bar{\mu}_{3,t}^+ - \bar{\mu}_{3,t}^- \quad (54)$$

Note the sign flip on l_3 because $C_{3c} = -1$.

Uncongested triangle. If all lines are uncongested ($\bar{\mu}_{l,t}^\pm = 0$):

$$\lambda_{A,t} - \lambda_{B,t} = \kappa_t x_1 \quad (55)$$

$$\lambda_{B,t} - \lambda_{C,t} = \kappa_t x_2 \quad (56)$$

$$\lambda_{A,t} - \lambda_{C,t} = -\kappa_t x_3 \quad (57)$$

Adding the first two: $\lambda_{A,t} - \lambda_{C,t} = \kappa_t(x_1 + x_2)$. From the third: $\lambda_{A,t} - \lambda_{C,t} = -\kappa_t x_3$. For consistency: $\kappa_t(x_1 + x_2 + x_3) = 0$, so $\kappa_t = 0$.

Key result for an uncongested meshed network: when no lines are congested, $\kappa_t = 0$ and all nodal prices are equal ($\lambda_{A,t} = \lambda_{B,t} = \lambda_{C,t}$). The physics enforced by the KVL constraint leads to price equalisation when there is no transmission scarcity.

Congested line l_1 . If $f_{1,t} = F_1$ ($\bar{\mu}_{1,t}^+ > 0$) while l_2 and l_3 are uncongested:

$$\lambda_{A,t} - \lambda_{B,t} = \kappa_t x_1 + \bar{\mu}_{1,t}^+ \quad (58)$$

$$\lambda_{B,t} - \lambda_{C,t} = \kappa_t x_2 \quad (59)$$

$$\lambda_{A,t} - \lambda_{C,t} = -\kappa_t x_3 \quad (60)$$

Now the price differences are non-zero and the KVL dual $\kappa_t \neq 0$ in general. The congestion on l_1 propagates through the physics to create price differences at all buses.

Transmission investment equilibrium. From (16), the capital cost of each line is recovered through congestion rents accumulated over all congested snapshots. Lines that are never congested at optimum will not be invested (or are already sized beyond their binding level).

8.3 Storage Investment with Cycling Constraint

8.3.1 Setup

Single node, two generators (baseload 1 and peaker 2), and a battery storage unit with extendable power capacity G^s and energy capacity $E_{\max}^s$.

The cycling constraint (CYC) requires $e_0 = e_T$. In practice this means the battery cannot start full and end empty (or vice versa) over the planning horizon. For annual planning, this enforces that the battery is in the same state at the end of the year as it was at the beginning.

8.3.2 KKT Conditions

Charge/discharge arbitrage. The shadow price ξ_t of stored energy propagates via (30). When storage cycles between a cheap hour t^+ and expensive hour t^- :

- At t^+ (charging, neither bound binding): $w_{t^+} \lambda_{t^+} = \xi_{t^+} / \eta^+$
- At t^- (discharging, neither bound binding): $w_{t^-} \lambda_{t^-} = \xi_{t^-} \eta^-$
- Over a full cycle: $\lambda_{t^-} / \lambda_{t^+} \geq 1 / (\eta^+ \eta^-)$

Investment equilibrium. From (33) and (34), power capacity is sized to the total hours when the charge or discharge rate is binding; energy capacity is sized to the total hours when the battery is full.

Effect of the cycling constraint. The cycling constraint has a dual variable ξ_{cyc} (the shadow price of the constraint $e_0 = e_T$). This initialises the chain of ξ_t values: the cycling dual anchors the intertemporal price chain so that the storage correctly values its energy across the full year. Without this constraint, the terminal condition would be free and storage could artificially end the year depleted, creating incorrect price signals.

Economic interpretation. A battery optimally sized will:

1. Earn enough arbitrage profit (charge cheap, discharge expensive) to recover power capital cost c^P through power rents
2. Earn enough full-reservoir rent to recover energy capital cost c^E
3. The cycling constraint ensures the arbitrage value is computed over a closed annual cycle, consistent with the annual capital cost annualisation

8.4 Multi-carrier Link: One Input, Two Outputs

8.4.1 Setup

A **multi-carrier conversion link** with one primary input and two outputs — representing for example a CHP plant (electricity + heat), a Sabatier reactor (electricity + CO₂ → methane + oxygen), or an electrolyser with heat recovery:

- Withdraws $p_{0,t}$ from input bus b_0 (price $\lambda_{0,t}$, e.g. electricity)
- Injects $\eta_1 p_{0,t}$ into output bus b_1 (price $\lambda_{1,t}$, e.g. methane); **efficiency** = $\eta_1 > 0$
- Injects $\eta_2 p_{0,t}$ into output bus b_2 (price $\lambda_{2,t}$, e.g. heat); **efficiency2** = $\eta_2 > 0$

Note: in PyPSA, a negative efficiency means the link *withdraws* from that bus (e.g. consuming CO₂). Here both $\eta_1, \eta_2 > 0$ so both are outputs.

The nodal balances receive:

$$b_0 : \quad \dots - p_{0,t} + \dots = d_{0,t} \quad (61)$$

$$b_1 : \quad \dots + \eta_1 p_{0,t} + \dots = d_{1,t} \quad (62)$$

$$b_2 : \quad \dots + \eta_2 p_{0,t} + \dots = d_{2,t} \quad (63)$$

Dispatch constraint: $0 \leq p_{0,t} \leq G_s$, capital cost c_s .

8.4.2 Stationarity Condition

Taking $\partial \mathcal{L} / \partial p_{0,t} = 0$:

$$\boxed{w_t o_s - \lambda_{0,t} + \eta_1 \lambda_{1,t} + \eta_2 \lambda_{2,t} - \underline{\mu}_{s,t} + \bar{\mu}_{s,t} = 0} \quad (64)$$

Rearranged as a value equation:

$$\underbrace{\eta_1 \lambda_{1,t} + \eta_2 \lambda_{2,t}}_{\text{value of outputs}} = \underbrace{\lambda_{0,t}}_{\text{cost of input}} - \underbrace{w_t o_s}_{\text{variable cost}} + \underbrace{\bar{\mu}_{s,t} - \underline{\mu}_{s,t}}_{\text{rent}} \quad (65)$$

Zero-profit condition (when $0 < p_{0,t} < G_s$):

$$\eta_1 \lambda_{1,t} + \eta_2 \lambda_{2,t} = \lambda_{0,t} - w_t o_s \quad (66)$$

The combined value of all outputs equals the cost of the input minus variable cost. This is a multi-product generalisation of the zero-profit condition introduced in Section 4.3.

8.4.3 Capacity Stationarity

From (13), the capital cost of the link is recovered through the infra-marginal rents $\bar{\mu}_{s,t}$ earned when the link is at full capacity.

8.4.4 Economic Interpretation

- $\lambda_{0,t}$: opportunity cost of the input (e.g. electricity price)
- $\eta_1 \lambda_{1,t}, \eta_2 \lambda_{2,t}$: per-unit revenue from each output, valued at market prices
- $w_t o_s$: variable operating cost
- $\bar{\mu}_{s,t} > 0$: infra-marginal rent when at full capacity
- $\underline{\mu}_{s,t} > 0$: shadow cost when idle (link not competitive)

A reference to Section 8.5 where LCOP for multi-carrier links is derived in detail.

8.5 Levelized Cost of Production for Multi-carrier Links

8.5.1 Motivation

Multi-carrier links model industrial processes that convert one carrier into (possibly multiple) products. The key economic question is: what is the *full* cost of producing one unit of each product, and how does it compare to the market price? This section derives the Levelized Cost of Production (LCOP) for such links and connects it to the KKT multipliers.

8.5.2 LCOP for a Multi-output Link

The LCOP for the primary product (output at b_1 , quantity $\eta_1 p_{0,t}$) requires attributing costs between outputs. Define the total annual cost:

$$\text{Total Cost} = c_s G_s + \sum_t (w_t o_s p_{0,t} + \lambda_{0,t} p_{0,t}) \quad (67)$$

Input flows are valued at their nodal KKT prices $\lambda_{0,t}$, reflecting the true opportunity cost of the input to the system. The full LCOP for the combined output bundle, normalised to unit of primary input, is:

$$\text{LCOP}_{\text{combined}} = \frac{c_s G_s + \sum_t w_t (o_s + \lambda_{0,t}/w_t) p_{0,t}}{\sum_t w_t p_{0,t}} \quad (68)$$

8.5.3 Allocating LCOP to Individual Products

When the link produces two outputs with prices $\lambda_{1,t}$ and $\lambda_{2,t}$, the revenue from output 2 acts as a *cost credit* for output 1. The LCOP for the primary product at b_1 is:

$$\text{LCOP}_{b_1} = \frac{c_s G_s + \sum_t (w_t o_s p_{0,t} + \lambda_{0,t} p_{0,t} - \eta_2 \lambda_{2,t} p_{0,t})}{\sum_t \eta_1 w_t p_{0,t}} \quad (69)$$

The numerator subtracts the value of the co-product at b_2 ; the denominator counts only the primary output (in MWh). This follows standard multi-product accounting: the secondary product's market value offsets the production cost attributed to the primary product.

8.5.4 LCOP Equals Product-Weighted Nodal Price

By the same proof as in Section 8.1.3, using the investment equilibrium (13) and dispatch stationarity (64):

$$\text{LCOP}_{b_1} = \frac{\sum_t w_t (\eta_1 p_{0,t}) \pi_{1,t}}{\sum_t w_t \eta_1 p_{0,t}} \equiv \bar{\pi}_{1,s}^{\text{prod}} \quad (70)$$

The LCOP of the primary product equals the product-weighted nodal price at the output bus b_1 , averaged over production hours — where the weight at each snapshot is the quantity of primary product delivered, $w_t \eta_1 p_{0,t}$.

This holds because of the interplay between two KKT conditions. The investment equilibrium (13) requires $c_s = \sum_t \bar{\mu}_{s,t}$: capital cost is recovered entirely through the infra-marginal rents $\bar{\mu}_{s,t}$ earned at hours when the link is at full capacity. At those same hours, complementary slackness (18) forces $g_{s,t} = G_s$, and the dispatch stationarity (64) gives $\bar{\mu}_{s,t} = \eta_1 \lambda_{1,t} + \eta_2 \lambda_{2,t} - \lambda_{0,t} + w_t o_s$, i.e. the rent is exactly the gap between the output value and the full production cost at that snapshot. Substituting both into the LCOP numerator collapses the capital cost term into the price signal $\pi_{1,t}$, yielding the product-weighted average.

8.5.5 Competing Links on the Same Bus

This situation is structurally identical to the baseload/peaker competition of Section 8.1: two technologies serving the same product bus coexist at long-run optimum because they occupy **different time niches**. Technology A (e.g. electrolysis) is cheap when its input is cheap (low electricity price); technology B (e.g. SMR) is cheap when A 's input is expensive. Neither can replace the other at all hours.

Suppose two links A and B (e.g. two hydrogen production technologies) both deliver their output to bus b_1 . At long-run optimum with both $G_A > 0$ and $G_B > 0$, the individual zero-profit condition holds for each:

$$\text{LCOP}_A = \bar{\pi}_{1,A}^{\text{prod}}, \quad \text{LCOP}_B = \bar{\pi}_{1,B}^{\text{prod}} \quad (71)$$

These two LCOPs are **generally not equal to each other**: each is the product-weighted average of $\pi_{1,t}$ taken over its *own* production hours. Technology A produces mainly when prices are low (setting the price itself as the marginal unit), while B produces mainly when prices are higher. Their respective product-weighted averages reflect these different price environments, so $\text{LCOP}_A \neq \text{LCOP}_B$ in general — just as the baseload and peaker LCOPs differ in Section 8.1.

What *is* common: both individually break even over the year. Neither earns supra-normal profit nor suffers a loss at long-run optimum. If either technology were unprofitable in its niche (its product-weighted average price fell below its LCOP), investment in it would be driven to zero.

LCOP vs. instantaneous shadow price $\pi_{1,t}$. The nodal price $\pi_{1,t}$ is a snapshot quantity; the LCOP is a lifecycle average. At any given snapshot, $\pi_{1,t}$ can be:

- **Equal to the SRMC of technology s :** s is the marginal unit ($\bar{\mu}_{s,t} = 0$, zero rent at that snapshot)
- **Above the SRMC of s :** s is at full capacity ($\bar{\mu}_{s,t} > 0$, earning infra-marginal rent)
- **Irrelevant:** s is idle ($\bar{\mu}_{s,t} < 0$)

The zero-profit condition says precisely that these snapshot deviations average out to zero over the year: $\text{LCOP}_s = \text{product-weighted average of } \pi_{1,t}$. The capital cost is recovered through the *accumulation* of infra-marginal rents at binding hours (13) — not from any single snapshot.

Practical computation in PyPSA. For link s delivering to bus b_1 :

$$\text{LCOP}_s = \frac{c_s G_s + \sum_t w_t (o_s + \pi_{0,t}) p_{0,t} - \sum_t w_t \eta_2 \pi_{2,t} p_{0,t}}{\sum_t w_t \eta_1 p_{0,t}} \quad (72)$$

where $\pi_{i,t} = \lambda_{i,t}/w_t$ is the nodal price in €/MWh.

8.6 CO₂ Emission Cap via Store

8.6.1 Setup

In PyPSA-Eur, the CO₂ emission cap is implemented as a **physical store component** on a dedicated CO₂ bus.

- A **CO₂ bus** b_{CO_2} with nodal price $\lambda_{\text{CO}_2,t}$
- A **CO₂ store** with energy capacity $E_{\text{max}}^{\text{CO}_2}$ (the emission budget in tCO₂), non-cyclic state of charge, zero cost
- Every emitting generator or link s with emission intensity ρ_s (tCO₂/MWh_{th}) and efficiency η_s injects $(\rho_s/\eta_s) g_{s,t}$ tCO₂ into b_{CO_2} per snapshot

The CO₂ store accumulates emissions:

$$e_{t+1}^{\text{CO}_2} = e_t^{\text{CO}_2} + \sum_s w_t \frac{\rho_s}{\eta_s} g_{s,t} \quad \forall t \quad (73)$$

with capacity constraint:

$$0 \leq e_t^{\text{CO}_2} \leq E_{\text{max}}^{\text{CO}_2} \quad \longleftrightarrow \quad \bar{\gamma}_t \geq 0 \quad (74)$$

8.6.2 Modified Stationarity for Emitting Generators

$$\boxed{w_t \left(o_s + \frac{\rho_s}{\eta_s} \lambda_{\text{CO}_2,t} \right) - \lambda_{n(s),t} - \underline{\mu}_{s,t} + \bar{\mu}_{s,t} = 0} \quad (75)$$

The CO₂ price $\lambda_{\text{CO}_2,t}$ augments the effective marginal cost of each emitting generator:

$$\lambda_{n(s),t} = \underbrace{w_t \left(o_s + \frac{\rho_s}{\eta_s} \lambda_{\text{CO}_2,t} \right)}_{\text{effective marginal cost}} + \bar{\mu}_{s,t} - \underline{\mu}_{s,t} \quad (76)$$

8.6.3 KKT Multipliers of the CO₂ Store

The intertemporal propagation of the carbon price follows equation (30) applied to the CO₂ store:

$$\lambda_{\text{CO}_2,t} = \lambda_{\text{CO}_2,t+1} + \bar{\gamma}_t \quad (77)$$

- $\bar{\gamma}_t > 0$ only when $e_t^{\text{CO}_2} = E_{\text{max}}^{\text{CO}_2}$ (budget fully exhausted): the system must curtail emitting generators or switch to zero-emission alternatives
- $\bar{\gamma}_t = 0$ when the store has remaining capacity
- $\lambda_{\text{CO}_2,t}$ is **non-increasing over time**: emitting early consumes the budget and reduces future flexibility

Summing (77) forward:

$$\lambda_{\text{CO}_2,t} = \sum_{\tau \geq t} \bar{\gamma}_\tau \quad (78)$$

The carbon price at t equals the sum of all future store-capacity rents. If the budget is never exhausted ($\bar{\gamma}_t = 0$ for all t), the carbon price is flat (constant over time). A tight budget produces a declining (or step-down) carbon price profile.

8.6.4 Comparison with a Global Constraint

An alternative to the store approach is a **global constraint** (e.g. PyPSA's `GlobalConstraint`):

$$\sum_t w_t \sum_s \frac{\rho_s}{\eta_s} g_{s,t} \leq E_{\max}^{\text{CO}_2} \quad (79)$$

This has a single dual variable $\mu^{\text{CO}_2} \geq 0$ — a constant carbon price applied identically at every snapshot.

Shadow price relationship. By the envelope theorem, both formulations attach a shadow price to the same parameter $E_{\max}^{\text{CO}_2}$. For the global constraint: $\partial \text{Cost}^* / \partial E_{\max}^{\text{CO}_2} = -\mu^{\text{CO}_2}$. For the store, relaxing $E_{\max}^{\text{CO}_2}$ relaxes every snapshot capacity constraint simultaneously, so the same derivative equals $-\sum_t \bar{\gamma}_t$. Using (78), this is $-\lambda_{\text{CO}_2, t=1}$ (the CO_2 price at the first snapshot). Therefore:

$$\mu^{\text{CO}_2} = \sum_t \bar{\gamma}_t = \lambda_{\text{CO}_2, t=1} \quad (80)$$

Are the two formulations actually equivalent in practice? In a standard perfect-foresight annual optimisation **without CCS**, the CO_2 store only ever fills: emissions are non-negative at every snapshot, so the state $e_t^{\text{CO}_2}$ increases monotonically from zero. The constraint $e_t^{\text{CO}_2} \leq E_{\max}^{\text{CO}_2}$ is therefore slack at all intermediate snapshots and only binds at the final one ($t = T$). Consequently $\bar{\gamma}_t = 0$ for $t < T$ and

$$\lambda_{\text{CO}_2, t} = \bar{\gamma}_T = \text{const.} = \mu^{\text{CO}_2} \quad \forall t \quad (81)$$

The carbon price is **constant and the two formulations are equivalent**.

When do they differ? The CO_2 store gives a time-varying price only when net CO_2 *removals* occur at some snapshots — i.e. when **CCS** is present and connected to the CO_2 bus with negative efficiency. CCS can draw the state back down from $E_{\max}^{\text{CO}_2}$, allowing the store to be full at one snapshot, partially drained later, then fill again. This creates multiple binding capacity constraints at different times ($\bar{\gamma}_t > 0$ for several t) and a declining price profile $\lambda_{\text{CO}_2, t}$. The global constraint cannot represent this dynamic and forces a flat price regardless of the CCS dispatch profile.

In PyPSA-Eur.

- $\lambda_{\text{CO}_2, t}$: `n.buses_t.marginal_price["co2"]` in €/t CO_2
- $\bar{\gamma}_t$: `n.stores_t.mu_upper["co2_store"]`
- For the global constraint: `n.global_constraints.mu["CO2Limit"]`

9 Summary: KKT Conditions

Central insight.

In a long-run competitive equilibrium, every investment is exactly justified by the rents it earns at binding hours. The KKT multipliers are simultaneously prices (for dispatch) and investment signals (for capacity). The LCOP of any optimally invested technology equals the energy-weighted average nodal price at its output bus over its production hours.

10 PyPSA Best Practices: Saving KKT Multipliers

After solving a PyPSA network (e.g. with linopy via `n.optimize()`), not all KKT multipliers are automatically stored in the network object. This section documents how to access and save the most relevant ones.

Variable	KKT condition	Economic meaning	Ref.
$g_{s,t}$ (generator)	$\lambda_{n,t} = w_t o_s + \bar{\mu}_{s,t} - \underline{\mu}_{s,t}$	Price = var. cost $\pm$ rent	(12)
G_s (generator cap.)	$c_s = \sum_t \bar{\mu}_{s,t}$	CAPEX = infra-marginal rents	(13)
$f_{l,t}$ (line flow)	$\lambda_{n^-,t} - \lambda_{n^+,t} = \sum_c \kappa_{c,t} C_{lc} x_l$ $+ \bar{\mu}_{l,t}^+ - \bar{\mu}_{l,t}^-$	Price diff = KVL coupling + congestion	(14)
F_l (line cap.)	$c_l = \sum_t (\bar{\mu}_{l,t}^+ + \bar{\mu}_{l,t}^-)$	Line CAPEX = congestion rents	(16)
$g_{s,t}$ (link/conv.)	$\lambda_{n^-,t} = \eta_s \lambda_{n^+,t} + w_t o_s + \bar{\mu}_{s,t}$	Input price = eff.-weighted output + cost + rent	(11)
G_s (link cap.)	$c_s = \sum_t \bar{\mu}_{s,t}$	CAPEX = conversion rents	(13)
$g_{k,t}^+$ (charging)	$w_t \eta^+ \lambda_{n,t} = \xi_t + \bar{\mu}_{k,t}^+ - \underline{\mu}_{k,t}^+$	Charge value = shadow energy price	(28)
$g_{k,t}^-$ (discharge)	$w_t \lambda_{n,t} / \eta^- = \xi_t - \bar{\mu}_{k,t}^- + \underline{\mu}_{k,t}^-$	Discharge value = shadow energy price	(29)
G_k^s (power cap.)	$c_k^P = \sum_t (\bar{\mu}_{k,t}^+ + \bar{\mu}_{k,t}^-)$	Power CAPEX = power rents	(33)
$E_{k,\max}^s$ (energy cap.)	$c_k^E = \sum_t \bar{\gamma}_{k,t}$	Energy CAPEX = full-reservoir rents	(34)
$p_{0,t}$ (multi-link)	$\eta_1 \lambda_{1,t} + \eta_2 \lambda_{2,t} = \lambda_{0,t} - w_t o_s + \bar{\mu}_{s,t}$	Output value = input cost - var. cost + rent	(64)
CO ₂ store	$\lambda_{\text{CO}_2,t} = \sum_{\tau \geq t} \bar{\gamma}_{\tau}$	Carbon price = future budget rents	(78)
Emitting generator	$\lambda_{n,t} = w_t (o_s + \frac{\rho_s}{\eta_s} \lambda_{\text{CO}_2,t}) + \bar{\mu}_{s,t}$	Effective cost augmented by carbon price	(76)
LCOP (any comp.)	$\text{LCOP}_s = \bar{\pi}_{n,s}^{\text{energy}}$	LCOP = energy-weighted LMP over production hours	(42)

Table 2: KKT stationarity conditions for capacity and dispatch optimisation. Dual variables with bars ($\bar{\mu}$) are zero when the associated constraint is not binding, positive only when it is. $\pi_{n,t} = \lambda_{n,t}/w_t$ denotes the nodal price in €/MWh.

10.1 Automatically Saved Multipliers

The following are saved by default after `n.optimize()`:

- **Nodal marginal prices** $\lambda_{n,t}$: `n.buses_t.marginal_price` [€/MWh]
- **Generator dispatch duals**: `n.generators_t.mu_upper` ($\bar{\mu}_{s,t}$), `n.generators_t.mu_lower` ($\underline{\mu}_{s,t}$)
- **Line flow duals** (congestion rents): `n.lines_t.mu_upper` ($\bar{\mu}_{l,t}^+$), `n.lines_t.mu_lower` ($\bar{\mu}_{l,t}^-$)
- **Link dispatch duals**: `n.links_t.mu_upper` ($\bar{\mu}_{s,t}$), `n.links_t.mu_lower` ($\underline{\mu}_{s,t}$)
- **Store energy duals**: `n.stores_t.mu_upper` ($\bar{\gamma}_{k,t}$), `n.stores_t.mu_lower` ($\underline{\gamma}_{k,t}$)
- **StorageUnit duals**: `n.storage_units_t.mu_upper`, `n.storage_units_t.mu_lower`

10.2 Capacity KKT Multipliers (Extendable Components)

The capacity duals ($\bar{\mu}_s^{p-\text{nom}}$ for generators, $\bar{\mu}_l^{s-\text{nom}}$ for lines, etc.) are **not saved automatically**. They must be extracted from the linopy model after solving.

Listing 1: Extracting capacity KKT multipliers from linopy

```

1 import pandas as pd
2
3 # After n.optimize()
4 m = n.model # the linopy Model object
5
6 # ---- Extendable generators: dual of p_nom constraint ----
7 # The constraint name depends on PyPSA version; typical names:
8 gen_cap_dual = (
9     m.constraints["Generator-ext-p_nom"]
10     .dual
11     .to_pandas()
12 )
13 # gen_cap_dual is a Series indexed by generator name
14 # Units: EUR/MW (= annualised CAPEX, by investment equilibrium)
15 n.generators.loc[gen_cap_dual.index, "mu_p_nom"] = gen_cap_dual
16
17 # ---- Extendable lines: dual of s_nom constraint ----
18 line_cap_dual = (
19     m.constraints["Line-ext-s_nom"]
20     .dual
21     .to_pandas()
22 )
23 n.lines.loc[line_cap_dual.index, "mu_s_nom"] = line_cap_dual
24
25 # ---- Extendable links: dual of p_nom constraint ----
26 link_cap_dual = (
27     m.constraints["Link-ext-p_nom"]
28     .dual
29     .to_pandas()
30 )
31 n.links.loc[link_cap_dual.index, "mu_p_nom"] = link_cap_dual
32
33 # ---- Storage: extendable energy capacity dual ----
34 store_e_dual = (
35     m.constraints["Store-ext-e_nom"]
36     .dual
37     .to_pandas()
38 )
39 n.stores.loc[store_e_dual.index, "mu_e_nom"] = store_e_dual

```

Verification: The investment equilibrium condition $c_s = \sum_t \bar{\mu}_{s,t}$ can be checked as:

Listing 2: Checking investment equilibrium

```

1 # For extendable generators
2 rents = n.generators_t.mu_upper.sum() # sum of infra-marginal rents
3 capex = n.generators.capital_cost     # annualised capital cost
4 # At optimum: rents[gen] ~= capex[gen] for all extendable generators
5 print((rents - capex)[n.generators.p_nom_extendable])

```

10.3 KVL Constraint Duals ($\kappa_{c,t}$)

The KVL constraint duals are accessible via:

Listing 3: Extracting KVL (Kirchhoff) constraint duals

```

1 # Kirchhoff (cycle) constraints
2 kvl_dual = (

```

```

3     m.constraints["Kirchhoff-Voltage-Law-s"]
4     .dual
5     .to_pandas()
6 )
7 # Rows = snapshots, columns = cycle indices
8 # Units: EUR/(MW*Ohm) or consistent with the LP formulation

```

The exact constraint name may vary with PyPSA version; use `m.constraints` to list all available constraints.

10.4 Cycling Store Constraint Dual

For a store with the cyclic state-of-charge condition (CYC):

Listing 4: Cycling store constraint dual

```

1 # The cycling constraint dual is typically part of the SOE
2 # initialisation. In linopy-based PyPSA:
3 cycling_dual = (
4     m.constraints["Store-cyclic-e"]
5     .dual
6     .to_pandas()
7 )
8 # This is the dual of  $e_0 = e_T$  (scalar per store)

```

10.5 Computing LCOP in PyPSA

Listing 5: Computing LCOP for a link delivering to bus b1

```

1 def compute_lcop(n, link_name, output_bus, output_eff_col="efficiency"):
2     """
3     Compute the LCOP for a PyPSA Link component.
4
5     Parameters
6     -----
7     n : pypsa.Network
8     link_name : str
9     output_bus : str (bus name of the primary output)
10    output_eff_col : str (column in n.links for output efficiency)
11    """
12    w = n.snapshot_weightings.generators # snapshot weights [hours]
13    p0 = n.links_t.p0[link_name] # input flow [MW]
14    eta_out = n.links.at[link_name, output_eff_col]
15
16    # Output energy [MWh]
17    output_energy = (w * eta_out * p0).sum()
18
19    # Capital cost [EUR/year]
20    capex = n.links.at[link_name, "capital_cost"] * \
21            n.links.at[link_name, "p_nom_opt"]
22
23    # Variable cost [EUR/year]
24    var_cost = (w * n.links.at[link_name, "marginal_cost"] * p0).sum()
25
26    # Input cost [EUR/year]
27    input_bus = n.links.at[link_name, "bus0"]
28    pi_in = n.buses_t.marginal_price[input_bus]
29    input_cost = (w * pi_in * p0).sum()

```



```

30
31     lcop = (capex + var_cost + input_cost) / output_energy
32
33     # Verification: should equal energy-weighted nodal price
34     pi_out = n.buses_t.marginal_price[output_bus]
35     output_flow = eta_out * p0
36     ewap = (w * output_flow * pi_out).sum() / (w * output_flow).sum()
37
38     return lcop, ewap # should be approximately equal at optimum

```

11 Nomenclature

Bars ($\bar{\cdot}$) denote upper-bound duals; underlines ($\underline{\cdot}$) denote lower-bound duals. Asterisks (*) denote optimal values. Subscripts follow the conventions: s for components, l for lines, k for stores, n for nodes, t for snapshots, c for cycles.

Sets and indices

s	Index for generators and dispatchable links
l	Index for passive transmission lines
k	Index for store (storage) components
n	Index for nodes (buses)
t	Index for time snapshots
c	Index for independent cycles in the network
$\mathcal{S}_n$	Set of all components injecting into node n
$\mathcal{K}_n$	Set of store components connected to node n
$\mathcal{L}_n^{\text{in}}, \mathcal{L}_n^{\text{out}}$	Sets of links with output at / input from node n
$\mathcal{P}_k$	k -th non-overlapping subset in a component partition

Primal decision variables

$g_{s,t}$	Dispatch of component s at snapshot t [MW]
G_s	Installed (optimised) capacity of component s [MW]
$f_{l,t}$	Power flow on line l at snapshot t [MW]
F_l	Capacity of line l [MW]
$e_{k,t}$	State of energy of store k at snapshot t [MWh]
$g_{k,t}^+, g_{k,t}^-$	Charge / discharge power of store k at snapshot t [MW]
G_k^s	Power capacity of store k [MW]
$E_{k,\text{max}}^s$	Energy capacity of store k [MWh]
$p_{0,t}$	Input flow of a multi-carrier link at snapshot t [MW]

Parameters

c_s, c_l	Annualised capital cost of component / line [€/MW/year]
c_k^P	Power capacity cost of store k [€/MW/year]
c_k^E	Energy capacity cost of store k [€/MWh/year]
o_s	Variable (marginal) operating cost of component s [€/MWh]
w_t	Snapshot weight: duration represented by snapshot t [hours]
$d_{n,t}$	Inelastic demand at node n at snapshot t [MW]
η_s	Conversion efficiency of component s [-]
η_k^+, η_k^-	Charge / discharge efficiency of store k [-]
η_1, η_2	Output efficiencies for the primary / secondary output of a multi-carrier link [-]
x_l	Reactance of line l [p.u.]
K_{nl}	Node–line incidence matrix (+1 if line ends at n , -1 if it starts there)
C_{lc}	Cycle–branch incidence matrix entry (± 1 or 0)
$n^-(s), n^+(s)$	Input / output bus of component s
ρ_s	CO ₂ emission intensity of component s [tCO ₂ /MWh _{th}]

Dual variables (KKT multipliers)

$\lambda_{n,t}$	Dual of the nodal power balance (KCL) at node n , snapshot t ; the Locational Marginal Price (LMP) . In [€/MW]; divide by w_t to obtain [€/MWh]
$\kappa_{c,t}$	Dual of the KVL cycle constraint for independent cycle c ; shadow price of Kirchhoff's voltage law
$\bar{\mu}_{s,t}$	Dual of the upper dispatch bound $g_{s,t} \leq G_s$; infra-marginal rent of component s at snapshot t [€/MW]
$\underline{\mu}_{s,t}$	Dual of the lower dispatch bound $g_{s,t} \geq 0$; shadow cost of idling [€/MW]
$\bar{\mu}_{l,t}^+, \bar{\mu}_{l,t}^-$	Duals of the upper / lower flow limits on line l ; congestion rents [€/MW]
$\xi_{k,t}$	Dual of the SOE evolution constraint for store k ; shadow price of stored energy [€/MWh]
$\bar{\gamma}_{k,t}$	Dual of the energy upper bound $e_{k,t} \leq E_{k,\max}^s$; full-reservoir rent [€/MWh]
$\underline{\gamma}_{k,t}$	Dual of the energy lower bound $e_{k,t} \geq 0$ [€/MWh]
$\bar{\mu}_{k,t}^+, \bar{\mu}_{k,t}^-$	Duals of the charge / discharge power upper bounds for store k [€/MW]
$\underline{\mu}_{k,t}^+, \underline{\mu}_{k,t}^-$	Duals of the charge / discharge power lower bounds for store k [€/MW]
$\lambda_{\text{CO}_2,t}$	Dual of the CO ₂ nodal balance; time-varying carbon price [€/tCO ₂]
$\bar{\gamma}_t$	Dual of the CO ₂ store energy upper bound; carbon budget scarcity rent [€/tCO ₂]
μ^{CO_2}	Dual of the global CO ₂ constraint (GlobalConstraint); constant carbon shadow price [€/tCO ₂]

Derived quantities

$\pi_{n,t}$	Nodal price at node n , snapshot t , defined as $\pi_{n,t} = \lambda_{n,t}/w_t$ [€/MWh]
LCOP_s	Levelized Cost of Production for component s [€/MWh]: total annualised cost divided by total energy produced
$\bar{\pi}_{n,s}^{\text{energy}}$	Production-weighted average nodal price for component s at bus n [€/MWh]; equals LCOP_s at LP optimum for extendable components
$\bar{\pi}_{b,s}^{\text{prod}}$	Same quantity normalised to primary-product output at bus b (used for multi-carrier links)
r_s	Total infra-marginal rent of fixed-capacity component s , $r_s = \sum_t \bar{\mu}_{s,t} G_s$ [€/year]
ρ_k	Cycling constraint dual for store k [€/MWh]; anchors the intertemporal price chain $\xi_{k,t}$

References

- [1] J. Hörsch, H. Ronellenfitsch, D. Witthaut, T. Brown, “Linear optimal power flow using cycle flows,” *Electric Power Systems Research*, vol. 158, pp. 126–135, 2018.
- [2] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge University Press, 2004. (Available free at stanford.edu/~boyd/cvxbook)
- [3] D. P. Bertsekas, *Nonlinear Programming*, 3rd ed. Athena Scientific, 2016.
- [4] S. Stoft, *Power System Economics: Designing Markets for Electricity*. Wiley-IEEE Press, 2002.
- [5] D. Kirschen and G. Strbac, *Fundamentals of Power System Economics*. Wiley, 2004.
- [6] H. R. Varian, *Microeconomic Analysis*, 3rd ed. W. W. Norton & Company, 1992.